AI PRODUCT MANAGEMENT CASE STUDY · FICTIONAL TARGET · OCTOBER 2026

Project Falcon

Multi-agent M&A due diligence, built in Claude Code without application code

An AI diligence team that researches an acquisition target across SEC filings, news, court records and financials, connects what no single source shows, and drafts an Investment Committee report, with people approving each stage.

4 fully blind graded runs	**97.4%** Critical and High recall, runs 1 to 3 average	**100%** recall on all 18 after the fixes (run 4)	**93.5%** recall on all 18, runs 1 to 3 average	**0** unsupported findings in any report

The problem

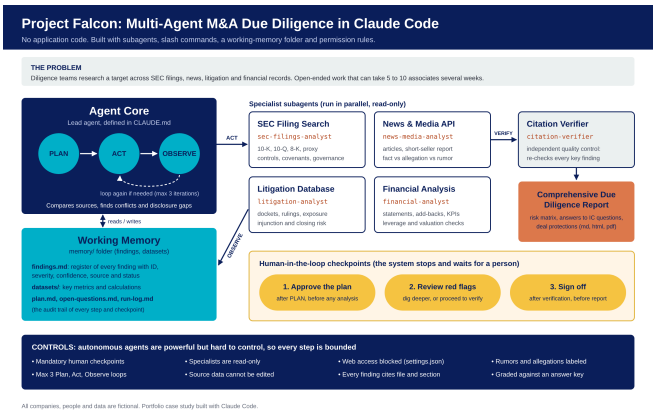
Diligence takes teams of associates weeks of reading filings, news, dockets and data-room financials, and the worst issues hide between sources. Autonomous agents could help, but they are the hardest AI pattern to control, so high-stakes work needs human checkpoints.

What I built

- **Agent Core:** a lead agent (`CLAUDE.md`) running a bounded Plan, Act, Observe loop (at most three iterations).
- **Four specialists in parallel:** SEC filings, news, litigation and financials. One source each, read-only.
- **Independent citation verifier:** re-opens every cited source and recomputes the numbers behind each Critical and High finding.
- **Working memory as plain files:** plan, findings register, conflicts, run log and every unedited agent report. Fully auditable.
- **Three human checkpoints:** approve the plan, review the red flags, sign off.
- **Testable guardrails:** no web access, protected sources, a prompt-injection rule, approval to open the answer key, 34 automated checks.

Key decisions

- **Specialists by source, not topic,** so connecting sources is the lead agent's job. The Critical issues only appear there.
- **Separation of duties:** analysts cannot change anything; a separate verifier checks them, like a deal-team reviewer.
- **Measured, not assumed:** 18 planted issues, 4 distractors and an answer key the analysts and verifiers never see.
- **Planned like a product:** specs and a phased plan with tests for each phase, agreed before building.



Architecture: Agent Core, four specialists, working memory, verifier and human checkpoints.

What the system concluded

All four blind runs: do not proceed at $24.00 per share. The committee decides.

1. The valuation used guidance the target had already replaced: about 18.3x on current guidance, not the 11.5x in the deal model.
2. A patent case turned against the target in July, with an injunction sought on robots behind about 42% of revenue. The next quarterly filing left the ruling out.
3. A customer worth 31% of revenue can leave after a change of control, and is building its own robots.

What I learned

- The verifier earned its place: 17 of 61 findings needed corrected wording before the report (one claimed 18 months of negative free cash flow; it was 12).
- The scorecard pointed to three instruction fixes: low-severity findings dropping out, a trend spread over five findings, and corrections applied late. I applied all three.
- Three runs, not one: the result held and the misses repeated, so they are design gaps, not luck. I fixed them; a fourth blind run found all 18.

How blind was the test? Each graded run started in a fresh session that never saw the answer key or an earlier run. An earlier sample run was only partly blind. Run 4 is one run with the fixes, so it shows no range yet. Next: quality fixes, then a second target.

In the repository: README, `runs/` (graded runs with scorecards), `examples/sample-run/` and `Project Memory/` (specs and plan).

Skills: AI product management (specs, phased plan, success metrics, evaluation, iteration from data), agentic workflow design and M&A domain depth. All companies, people and figures are fictional; not investment advice.